

Series and Progressions: Masterclass

Author: Senior TCS NQT Examiner (10+ Years Experience)

1. Concept Clearance: AP & GP

Arithmetic Progression (AP)

- **Definition:** Sequence where the difference between consecutive terms is constant (d).
- **N-th term:** $T_n = a + (n-1)d$
- **Sum of n terms:** $S_n = (n/2) * [2a + (n-1)d]$ OR $S_n = (n/2) * (a + l)$ where l is the last term.
- **Shortcuts:** If you know the sum of 3 terms, the middle term is $Sum / 3$.

Geometric Progression (GP)

- **Definition:** Sequence where the ratio between consecutive terms is constant (r).
- **N-th term:** $T_n = a * r^{n-1}$
- **Sum of n terms:** $S_n = a * (r^n - 1) / (r - 1)$ for $r > 1$.
- **Infinite GP Sum:** $S = a / (1 - r)$ where $|r| < 1$.

2. Examiner's Trap Zone

Trap 1: The "Add 1" Fence Post Error

If the question asks: "How many trees are planted every 5 meters in a 100m long line?", students often calculate $100/5 = 20$. But the answer is 21 (you need a tree at the 0m mark). Always check if you need to add 1 to your result for physical objects in a line.

Trap 2: GP vs AP Confusion

Check the difference vs. ratio. A sequence 2, 4, 8 is GP, not AP. Using AP formulas here is a common failure point.

3. 10 High-Priority Questions

Q1: Find the 10th term of AP: 2, 7, 12...

Sol: $a=2$, $d=5$, $n=10$. $T_{10} = 2 + (10-1)*5 = 2 + 45 = 47$.

Q2: Find the sum of the first 20 terms of AP: 5, 8, 11...

Sol: $a=5$, $d=3$, $n=20$. $S_{20} = (20/2) * [2(5) + 19(3)] = 10 * [10 + 57] = 670$.

Q3: Find the 5th term of GP: 3, 6, 12...

Sol: $a=3$, $r=2$, $n=5$. $T_5 = 3 * 2^4 = 3 * 16 = 48$.

Q4: Find sum of infinite GP: 1, 1/2, 1/4...

Sol: $a=1$, $r=1/2$. $S = 1 / (1 - 0.5) = 1 / 0.5 = 2$.

Q5: In an AP, if 3rd term is 7 and 7th term is 15, find 10th term.

Sol: $a+2d=7$, $a+6d=15$. Subtracting: $4d=8$, $d=2$. $a=3$. $T_{10} = 3 + 9(2) = 21$.

Q6: How many terms of AP 1, 3, 5... sum to 100?

Sol: $\text{Sum} = n^2$. $n^2 = 100 \Rightarrow n=10$.

Q7: Find the geometric mean of 4 and 16.

Sol: $GM = \sqrt{4*16} = \sqrt{64} = 8$.

Q8: A tree grows 10% taller each year. If it is 2m tall, how tall in 3 years?

Sol: $T_3 = 2 * (1.1)^3 = 2 * 1.331 = 2.662\text{m}$.

Q9: If the sum of first 3 terms of an AP is 24, find the middle term.

Sol: Middle term = $\text{Sum}/3 = 24/3 = 8$.

Q10: Find sum of all odd numbers between 1 and 50.

Sol: AP: 1, 3... 49. $n = (49-1)/2 + 1 = 25$. $\text{Sum} = (25/2)*(1+49) = 25*25 = 625$.